

Michael Cassidy

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EDUCATION

Queen's University

Sept 2022 – May 2027

Bachelor of Applied Sciences in Engineering, Mechatronics and Robotics

Kingston, ON

- Recipient of Schulich Leader Scholarship valued at \$80 000
- Queen's Engineering Competition - Senior Design - 1st Place
- GPA: 3.83, Dean's Scholar

EXTRACURRICULARS

Team Captain

Apr 2025 - Present

Queen's Aerospace Design Team

Kingston, ON

- Led a 50-person sub team responsible for the full design and construction of a fixed-wing aircraft for the AIAA competition, including CAD modeling, stress analysis, manufacturing, and systems integration.
- Oversaw mechanical subsystem designs through regular design review and inter-team meetings to ensure successful design integration and identify technical improvements.
- Developed aerodynamic models of the aircraft using OpenVSP and a custom Python script for dynamics calculations to optimize stability and efficiency.
- Fabricated the aircraft shell by performing carbon fibre layups with custom 3D-printed moulds.

Team Captain

Sept 2022 - May 2025

Queen's VEXU Robotics Team

Kingston, ON

- Led an 80-person technical team across mechanical, electrical and software subteams.
- Designed several types of gearboxes, including multistage planetary and cycloidal to transmit power with tight space requirements and used several types of gears including spur, helical, herringbone and bevel.
- Led the mechanical development of four robots over three years including motorized gear trains and pneumatic actuators, complex linkages, manipulators and power takeoff mechanisms.
- Competed at the VEXU Robotics World Championships every year on the team and won VEX AI Skills World Champion and the Build Award at worlds, recognizing durability, reliability, meticulous robot construction.
- Modeled and integrated manufacturable parts using SOLIDWORKS and OnShape, applied kinematic/dynamic analysis using MATLAB, and included design tolerance to meet performance and reliability requirements.
- Fabricated components using 3D-printing, machining, and moulding using a rapid prototyping and testing design philosophy to meet weight, strength, and space constraints.

PROFESSIONAL EXPERIENCE

Software Engineer Intern

May – Aug 2023, 2024, 2025

Teranet

Toronto, ON

- Led development of 12 Java REST APIs in 2024, reducing the average request response times by over a minute for key operations compared to the legacy system, significantly improving efficiency.
- Developed a new integrated application in 2025, cutting response times by over 100ms across all requests and reducing operating costs by unifying functionality from multiple legacy systems.
- Enhanced codebase maintainability and team collaboration by standardizing design practices and improving readability across applications.
- Recognized by management as a top-performing intern, demonstrating skills on par with established developers and earning the highest performance rating for a co-op student.

Robotics and Programming Instructor

Jul 2019 - Aug 2023

RoboEDU

Toronto, ON

- Taught robotics and programming classes to over 300 students from ages 4 to 16, clearly explaining engineering concepts in an engaging and age-appropriate way.
- Coached competitive robotics teams, guiding mechanism design, and teamwork, focusing on the design process.
- Developed lesson plans and projects that helped students build practical skills in robotics engineering.

SKILLS

Languages: English, French

Software Skills: SOLIDWORKS, OnShape, OpenVSP, FEA, Python, MATLAB, C++, Rust, Java, Excel

Manufacturing Methods: 3D Printing, Moulding, CNC Machining, Carbon Fibre Layups, Hand and Power Tools